HANAN HUSSAIN

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Computer Engineering grad with hands-on experience building embedded systems, full-stack platforms, and AI-powered pipelines. Strong foundation in Python, C/C++, and hardware integration, with a track record of delivering complete systems from design through verification.

EDUCATION

University of Saskatchewan | Saskatoon, SK
Bachelor of Science in Computer Engineering | June 2026
Relevant Coursework: Software Engineering, Digital Design, Embedded Systems, Verification, Computer Networks

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Verilog, JavaScript (React, Node.js), HTML/CSS, SQL
Frameworks & Tools: Django, Next.js, Tailwind CSS, Linux, Bash, Git, MATLAB, Quartus, ModelSim
Hardware & Embedded: Raspberry Pi, Arduino, ARM Microcontrollers, FPGA (Intel), Logic Design, FPGA Programming
Soft Skills: Strong communication, teamwork, adaptability, attention to detail, time management, problem solving

PROJECTS

AI Code Review Pipeline | 2026
Python, CrewAI, Claude API, FastAPI, GitHub API, GCP Cloud Run, Docker

- Built a multi-agent AI code review system using CrewAI and Claude API that automatically reviews GitHub PRs via webhooks; deployed to **GCP Cloud Run** with Docker — serverless, scales to zero, permanent public URL
- Engineered **3 parallel AI agents** (security, performance, style) running concurrently via ThreadPoolExecutor, reducing review time by **3x vs sequential execution**
- Integrated GitHub Review API to post **inline comments pinned to specific lines of code**
- Implemented HMAC webhook signature verification, async background processing, and PR diff parsing to validate comment positions

Two-Way Portable English-to-ASL Translator (Capstone) | Sept 2025 - Apr 2026
Raspberry Pi 5, Python, MediaPipe, VOSK STT, C, Linux, I2S/I2C/CSI, Edge Inference

- Designed and integrated a real-time bidirectional ASL-English translation system on Raspberry Pi 5; achieved **end-to-end latency under 2 seconds** with fully offline operation across **100+ word vocabulary**
- Developed a key-point-based gesture recognition pipeline using Python and MediaPipe; achieved **>85% classification accuracy** across gesture classes through landmark extraction, inference logic, and token mapping
- Architected a modular multi-threaded system managing **5 concurrent subsystems** (camera, microphone, display, speaker, control) on a single embedded platform within a **$500 hardware budget**
- Implemented speech-to-text (VOSK) and text-to-speech pipelines with grammar conversion and ASL token sequencing
- Performed system-level verification across **4-person team (~11 hrs/week per member)** including timing analysis, latency benchmarking, functional validation, and integration debugging over a **7-month development cycle**

KawaKraft - Full-Stack E-Commerce Platform | 2025
Django, Next.js, Tailwind CSS, PostgreSQL, REST API, GitHub

- Designed and launched a full-stack e-commerce platform with **3+ secure REST API endpoints** covering product management, authentication, and customer orders; deployed with version-controlled CI workflow via GitHub
- Built responsive UI/UX with Next.js and Tailwind CSS, achieving **100% mobile-responsive layouts** across all screen breakpoints

8-bit Microprocessor (Verilog HDL) | 2024
Verilog, ModelSim, Quartus, Intel FPGA

- Designed a modular 8-bit microprocessor in Verilog with **3 core components** (sequencer, instruction decoder, ALU); synthesized and validated on Intel FPGA using Quartus with **zero critical timing violations** in ModelSim simulation

CERTIFICATIONS

- Google Cloud Fundamentals - Core Infrastructure (2025)
- GitHub Foundations Certificate (2024)
- AWS Educate - Cloud Computing (2023)

VOLUNTEER & ACTIVITIES

- University of Saskatchewan Engineering Society - Events Team | Assisted organizing community outreach and networking events for engineering students